Question	Answer	Marks	Partial Marks	Guidance
12(a)	$\frac{13x-4}{24}$ final answer	**3**	**M2** for $\frac{6x+9x-2x-4}{24}$ oe or **M1** for $\frac{6x+3(3x)-2(x+2)}{24}$ oe	For M2 and M1 accept equivalent fractions over a common denominator, accept as 3 separate fractions For M1 condone 1 error or allow just for numerator
12(b)	$(3x-y)(a+4y)$ final answer	**1**		Condone omission of final bracket
13	$\frac{3}{2}$ oe cao	**4**	**B3** for $16r^2 = 36$ or better or **M2** for $\pi r^2 16 = \frac{4}{3}\pi 3^3$ or **M1** for $\pi r^2 16$ or $\frac{4}{3}\pi 3^3$	Allow 3.14 etc for method
14(a)	70 **and** opposite angles of a cyclic quadrilateral sum to 180 oe	**2**	**B1** for 70 or for fully correct reason	Accept e.g. cyclic quadrilateral with no incorrect statement or angle Accept angles in opposite segments are supplementary See AG
14(b)(i)	65	**2**	**FT** 180 – 45 – *their* 70 for 2 marks **B1** for angle $BDC = 45$	May be marked on diagram Consider other angles for B1 at STM.
14(b)(ii)	65	**1**	**FT** *their* (b)(i)	
15(a)	(1, 13)	**2**	**B1** for one correct coordinate	
15(b)	$\begin{pmatrix}-9\\3\end{pmatrix}$	**2**	**B1** for $\begin{pmatrix}-9\\k\end{pmatrix}$ or $\begin{pmatrix}k\\3\end{pmatrix}$	